

References: Read Reactor Theory Manual Chapter 1: §1.5**Reference the following content on the class Moodle page:**

- *Lecture Module #5 – Lecture Slides*
- *Lab Module #4 – 1/M Approach to Criticality PDF & Data File*

1. Sub-Critical Multiplication: Mark the Following Statements **TRUE** or **FALSE** (Assume $K_{eff} < 1$ for all cases):

- _____ a) Equal changes in K_{eff} do NOT result in equal changes in equilibrium neutron population N_{eq} .
- _____ b) As K_{eff} approaches 1.0, more generations are required to reach equilibrium; therefore, less time is required to reach equilibrium.
- _____ c) If K_{eff} is increased halfway to 1.0 from its initial value, the neutron count rate will decrease by half.
- _____ d) As K_{eff} approaches 1.0, the # of neutrons lost from the fission chain decreases (i.e. $N_i(1-K_{eff})$ decreases)

2. Calculate changes in K_{eff} from step additions in reactivity and each corresponding equilibrium countrate (CR). Sketch what the resultant trace would look like on the logarithmic NFM plot. Use the reactivity additions found in Chapter 1, problem #2 on page 19 of Reactor Theory Manual (use a separate piece of paper, please do not cram into this small section):
Note: $K_{eff} = 1/(1-\rho)$; $CR_2 = CR_1 \times (1-K_1)/(1-K_2)$.

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1/M Lab Questions - Fill in the blanks:

REF: Lab #4 – 1/M Approach to Criticality PDF & Excel Data Plot (from lab)

1/M Startup - Fill in the blanks:

1. During a 1/M startup, do the following values increase ($\uparrow$), decrease ($\downarrow$) or stay the same ($\leftrightarrow$)?:
S ____ N_f ____ CR_o ____ CR_i ____
2. During a 1/M startup, what limits are approached for the following values? (0, 1, or ∞):
 $K_{eff} \rightarrow$ ____ $M \rightarrow$ ____ $1/M \rightarrow$ ____
3. Which detector in the reactor core is the Neutron Flux Monitor (NFM) connected to?
4. What are the two conditions used (*i.e. whichever happens first*) when determining how far to withdraw the control rods for each 1/M step?
5. How many times in lab did we determine how far to withdrawal the rods by doubling the count rate?
6. Approximately what critical gang rod height (in inches) did our 1/M plot converge to?
7. Apart from the first rod pull, was our plotted 1/M approach to criticality conservative or non-conservative?